

CRISTOFF STAN

Senior Software Engineer – Python, AI/ML, LLM Engineering

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PROFESSIONAL SUMMARY

Senior Software Engineer with 10+ years delivering AI-enabled B2B SaaS, e-commerce, and enterprise platforms using **Python 3.6–3.12, FastAPI, Django, PyTorch, TensorFlow, scikit-learn, PostgreSQL, and AWS/Azure**; specializes in production ML pipelines, RAG and LLM applications, model evaluation, MLOps, and cloud-native architecture that turn complex data into reliable automation, faster decisions, and measurable operational value.

TECHNICAL SKILLS

- **Python and Backend Engineering:** Python 3.8–3.12, FastAPI, Django, Django REST Framework, Flask, Pydantic, SQLAlchemy, Alembic, Celery, asyncio, Unicorn, Uvicorn, REST APIs, GraphQL, WebSockets, background workers and scheduled processing.
- **Machine Learning and Deep Learning:** PyTorch 1.x–2.x, TensorFlow 2.x, Keras, scikit-learn 1.x, XGBoost, LightGBM, pandas, NumPy, SciPy, feature engineering, supervised learning, anomaly detection, recommendation systems, forecasting and model calibration.
- **Generative AI and LLM Engineering:** OpenAI and Azure OpenAI APIs, Anthropic Claude, Hugging Face Transformers, LangChain, LlamaIndex, retrieval-augmented generation, embeddings, semantic search, hybrid retrieval, reranking, function calling, structured outputs, prompt engineering and AI agents.
- **LLM Evaluation and Safety:** Ragas, DeepEval, prompt evaluation datasets, groundedness testing, hallucination detection, human-in-the-loop review, PII redaction, prompt injection controls, content filtering, token optimization and model fallback strategies.
- **Data and Vector Platforms:** PostgreSQL, MySQL, MongoDB, DynamoDB, Redis, Snowflake, pgvector, Pinecone, FAISS, Elasticsearch/OpenSearch, data modeling, schema evolution, SQL optimization, indexing and distributed caching.
- **MLOps and Data Engineering:** MLflow, Kubeflow, Apache Airflow, dbt, Feast, SageMaker, Azure Machine Learning, model registries, feature stores, experiment tracking, batch inference, online inference, drift monitoring and reproducible pipelines.
- **Cloud and DevOps:** AWS Lambda, ECS, EKS, SageMaker, Bedrock, API Gateway, S3, RDS, SQS, SNS and CloudWatch; Azure AI Services, Azure Machine Learning, Functions, AKS, Service Bus, Key Vault and Application Insights.
- **Infrastructure and Delivery:** Docker, Kubernetes, Terraform, Helm, GitHub Actions, GitLab CI, Jenkins, SonarQube, container scanning, infrastructure as code, automated model validation, canary releases and rollback automation.
- **Testing and Observability:** pytest, unittest, Hypothesis, Factory Boy, Testcontainers, Locust, Pact, Postman, Prometheus, Grafana, OpenTelemetry, Datadog, New Relic, Sentry, structured logging and distributed tracing.
- **Architecture and Leadership:** Microservices, event-driven architecture, modular monoliths, Domain-Driven Design, transactional outbox, idempotency, OAuth 2.0, OIDC, JWT, RBAC, architecture reviews, mentoring, technical planning and production incident leadership.

WORK EXPERIENCE

BEVY

Senior Software Engineer

(September 2021 – June 2026)

Remote, Romania

- Architected multi-tenant AI platforms using **Python 3.8–3.12, FastAPI, Django, PostgreSQL, Redis, Kafka, and AWS**, defining service boundaries for document intelligence, recommendation, search, and workflow automation products.
- Designed **retrieval-augmented generation** services with **LangChain, LlamaIndex, pgvector, OpenSearch, and managed LLM APIs**, combining hybrid retrieval, metadata filters, reranking, and citation-aware response generation.
- Reduced average document-processing time by **34%** by replacing synchronous OCR and enrichment flows with **Celery workers, event-driven queues, batched model inference, and idempotent retry handling**.
- Built model-training pipelines with **PyTorch, scikit-learn, MLflow, Airflow, and versioned feature datasets**, supporting reproducible experiments, approval gates, and controlled promotion into production endpoints.
- Resolved hallucinated answers and retrieval misses caused by poor chunking, stale embeddings, and ambiguous prompts by introducing semantic chunking, evaluation datasets, prompt versioning, and grounded-answer thresholds.
- Led upgrades from **Python 3.8 to Python 3.11/3.12**, replacing incompatible libraries, improving type coverage with mypy and Pydantic, and validating behavior through regression and load testing.
- Implemented secure AI APIs with **OAuth 2.0, OIDC, tenant-aware RBAC, PII redaction, encrypted secrets, audit trails, and rate limits** before prompts reached external model providers.
- Created offline and online evaluation workflows using **Ragas, DeepEval, custom business metrics, human review queues, and canary releases** to compare answer quality, latency, token use, and task completion.
- Strengthened MLOps delivery with **Docker, Kubernetes, Terraform, GitHub Actions, SageMaker, Azure Machine Learning, Prometheus, and OpenTelemetry**, adding automated tests, model checks, and rollback controls.
- Mentored engineers and data scientists through architecture reviews, prompt and model debugging, incident retrospectives, and production-readiness assessments that improved ownership across AI-enabled services.

Syscom Digital

Software Engineer

(October 2019 – August 2021)

Remote, Romania

- Developed Python services and operational tools for digital commerce, customer support, and fraud-review workflows using **Python 3.7–3.9, Django, Flask, PostgreSQL, Redis, and AWS-managed infrastructure**.
- Built product-ranking, demand-forecasting, and anomaly-detection features with **pandas, scikit-learn, XGBoost, and scheduled Airflow pipelines**, exposing predictions through versioned REST endpoints.
- Reduced recurring payment-review false positives by **27%** by improving feature engineering, rebalancing training data, calibrating classification thresholds, and adding merchant-specific evaluation segments.
- Investigated inventory prediction drift caused by seasonal demand changes, incomplete event feeds, and delayed refunds, then introduced data-quality checks, retraining triggers, and controlled backfill procedures.

- Implemented asynchronous feature computation and batch inference with **Celery, RabbitMQ, Redis, and Amazon S3**, preventing long-running analytics jobs from blocking checkout and administration APIs.
- Created monitoring dashboards for prediction distributions, model confidence, missing features, API latency, and error rates using **CloudWatch, Grafana, structured logs, and automated alert thresholds**.
- Fixed schema mismatches between payment providers and internal models by introducing **Pydantic validation, canonical event contracts, dead-letter queues, and replayable transformation pipelines**.
- Expanded CI/CD with **Jenkins, GitLab CI, Docker, pytest, migration validation, and staged deployment checks** for application services, data pipelines, and model artifacts.
- Collaborated with product managers, analysts, QA engineers, and support teams to convert ambiguous operational problems into measurable experiments, production features, and monitored acceptance criteria.

Soft Build

Software Developer

(June 2016 – July 2019)

Remote, Romania

- Built Python applications and data-processing services with **Python 3.5–3.7, Django, Flask, PostgreSQL, MySQL, pandas, NumPy, and scikit-learn** for commerce, booking, reporting, and customer-management products.
- Improved forecast accuracy by **31%** for appointment demand and inventory planning by cleaning historical data, adding calendar and seasonality features, and comparing regression and tree-based models.
- Developed REST APIs for pricing, recommendation, reporting, and customer segmentation, separating business rules, model logic, persistence, and serialization into maintainable application layers.
- Resolved duplicated records, stale aggregates, and failed batch imports by adding transactional writes, idempotency keys, checkpointed jobs, and detailed rejection reports for invalid source data.
- Moved email classification, report generation, and nightly scoring into **Celery background workers**, improving request latency while adding retry policies and visibility into failed tasks.
- Created reusable dashboards and internal analysis tools that allowed operations teams to review predictions, override recommendations, and trace the input data behind automated decisions.
- Added **pytest, unittest, data fixtures, model-validation tests, and integration coverage** around pricing calculations, feature pipelines, authentication middleware, and critical API endpoints.
- Worked with senior engineers on SQL execution plans, indexing, Linux deployment scripts, model-performance reviews, and production incident fixes across data-intensive business applications.

Decima Digital

Software Developer

(November 2015 – April 2016)

Remote, Estonia

- Delivered Python and Django functionality using **Python 2.7/3.4 and Django 1.7** for content-driven websites, administration tools, reporting modules, and lightweight data-processing workflows backed by PostgreSQL and MySQL.
- Maintained server-rendered templates and legacy JavaScript screens while creating reusable Python utilities for text normalization, CSV imports, scheduled reports, and content categorization.
- Built Django views, service utilities, forms, and data-access components for content retrieval, file uploads, notification delivery, account workflows, and basic reporting features.
- Fixed production defects involving encoding failures, malformed CSV files, missing database values, broken validation, and unclear error messages by strengthening backend checks and recovery paths.
- Improved slow reporting operations by reviewing ORM queries, adding indexes, using bulk inserts, and replacing repeated record loops with targeted SQL and batched transformations.

- Created small rule-based and statistical text-classification utilities with **NLTK** and **scikit-learn** to route incoming content into review queues while preserving manual override controls.
- Participated in browser testing, Linux deployment preparation, smoke checks, data validation, and rollback planning, helping reduce release issues across several client applications.

Borna Technology

Software Developer Intern

(September 2014 – August 2015)

Tallinn, Estonia

- Assisted senior engineers with **Python 2.7/3.3**, **Django 1.5**, **JavaScript**, **PostgreSQL**, and **MySQL** tasks across internal tools, client websites, and basic data-driven business applications.
- Built reusable Django templates, validation scripts, and data-import helpers while learning structured debugging, browser compatibility handling, and disciplined Git collaboration.
- Supported backend development for Django views, form handlers, administration pages, CSV processing, and server-side validation using Python, Django ORM, and relational databases.
- Investigated layout defects, invalid form submissions, Python exceptions, and inconsistent JSON responses by reproducing issues and documenting reliable diagnostic steps for senior developers.
- Contributed to QA checks, deployment notes, simple test automation, and code reviews, establishing a strong foundation in maintainable software delivery and production support.

EDUCATION

Tallinn University

Bachelor's Degree in Computer Science,

September 2011 – May 2014

CERTIFICATIONS

- **AWS Certified Machine Learning Engineer – Associate**
- **AWS Certified Solutions Architect – Associate**
- **Microsoft Certified: Azure AI Engineer Associate**
- **Certified Kubernetes Application Developer — CKAD**
- **Professional Scrum Developer I — PSD I**

PROJECTS

Enterprise RAG Knowledge and Document Intelligence Platform

- Designed a secure enterprise assistant using **Python**, **FastAPI**, **LangChain**, **LlamaIndex**, **Azure OpenAI**, **pgvector**, **OpenSearch**, and **PostgreSQL** to retrieve information from contracts, policies, support records, and internal knowledge sources.
- Implemented semantic chunking, hybrid retrieval, cross-encoder reranking, metadata authorization, source citations, prompt versioning, PII redaction, response caching, and automated **Ragas** and **DeepEval** quality checks.
- Added human-review queues, groundedness thresholds, provider fallback, token-budget controls, and OpenTelemetry tracing to balance answer accuracy, security, response latency, and operating cost.

Predictive Operations and Anomaly Detection Platform

- Built an ML platform using **Python, PyTorch, scikit-learn, XGBoost, Airflow, MLflow, Kafka, PostgreSQL, and AWS SageMaker** for demand forecasting, transaction anomaly detection, and operational risk scoring.
- Created reproducible feature pipelines, time-aware validation, model registries, batch and online inference endpoints, drift monitoring, canary releases, and automated rollback when quality or latency crossed production thresholds.